R&D and application of whole-process control platform for engineering projects based on EPC and big data

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Abstract

The digital platform can realise the detailed management of project progress, quality, safety and resources, and provide three-dimensional guidance for project construction. By providing progress management function modules, it enables managers to track and supervise the progress of the project and identify and deal with problems that may arise in a timely manner. Through the quality control of the project, the quality indexes of each project can be managed in a standardised manner, and the quality of the project can be guaranteed to meet the predetermined requirements through testing and evaluation of the project. Through the safety management of the project, the safety norms and operation procedures of the EPC (Engineering Procurement Construction) project are formulated, and the hazards of the project are recorded and analysed, so as to ensure the safety of the project. With the sustained and steady growth of China's economy in recent years and the accelerating pace of urbanisation, the people's living standards continue to improve; at the same time, due to China's large population base, the per capita housing area is small, so that China's large and medium-sized cities, commercial housing, sheltered housing and public building demand remains high, the construction industry development is booming. Construction not only need to invest a lot of labour and machinery, there are a large number of building materials, semi-finished products production, transport, storage and supply work, to make people, materials, machinery and other factors of production in an orderly manner to play its role, the difficulty and complexity is increasing. Advanced and reasonable schedule management methods can enable construction enterprises to achieve the optimal balance of construction cost and construction period while realising project quality objectives'. Project schedule control is very important for the success or failure of project management, on the basis of meeting the requirements of project quality, the optimisation of cost and duration determines the success or failure of project management. Therefore, it is of far-reaching significance to explore the project schedule management techniques.

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CCS Concepts

• **Information systems** → World Wide Web; Web interfaces; Wikis.

Keywords

EPC and Big Data, Engineering project management, Management and control platform building, Platform Development

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1 Introduction

Under the traditional management mode, the emphasis on the whole process of project implementation from the overall integrated control, but the reality is that, because the integrity system of the whole society is not sound, the entities from the protection of their own interests or even the pursuit of profit maximisation, it is often easy to artificially form a variety of information silos, resulting in information asymmetry, thus affecting the synergy of the whole process of construction and precision, reducing production efficiency and project quality, and even causing major quality accidents or safety incidents. project quality, and even cause major quality or safety accidents. Establishing an intelligent management platform for the whole construction process based on the industrial Internet platform can effectively break the information islands and eliminate the information asymmetry, and it can make use of the core value of blockchain, which is 'establishing a consensus mechanism of mutual trust between entities that do not have sufficient trust', to upload the key data of the whole process of construction to achieve the transparency and intelligent management of the construction process. Intelligent management.

At present, a number of domestic organisations have launched intelligent management platforms for the whole construction process. For example, the intelligent construction information management platform researched and developed by China Construction First Bureau is applied to Shenzhen International Convention and Exhibition Centre. Through the 'Internet +' intelligent collaboration platform, it realizes intelligent management on 11 management systems, such as personnel and machinery positioning, material tracking and acceptance, progress control, quality inspection, TSP environment monitoring, etc., which constitutes the CPU of the entire intelligent construction site, making the whole construction

site transparent and intelligent. CPU, making the construction process of Shenzhen International Convention and Exhibition Centre project more intelligent and efficient. The comprehensive digital management platform for the whole process of infrastructure developed by State Grid has digital, intelligent and open sharing features, and serves the 'four-in-one' intelligent platform (digital management platform, infrastructure work platform, resource sharing platform and decision-making support platform) for all levels of grid infrastructure professions and related professional users, which realises the sharing of data of all infrastructure professions, and the sharing of data of all departments of the company. It realises the sharing of professional data of infrastructure, the business integration of various departments of the company and the synergistic development of the company and related industries. Guangzhou Pearl River Construction Group and Guangzhou Yuejian Sanhe Software Co., Ltd. jointly developed the 'Guangzhou Pearl River Construction Group Building Construction Intelligent Control Platform', which is an internet platform based on blockchain technology and serves Guangzhou Pearl River Construction Development Company Limited, its molecular companies, project departments and subcontracting units, as well as suppliers of materials and equipments. It realises the intelligent management of the whole industrial chain from project bidding, contract evaluation, task issuance, material control to project settlement.

Engineering project management is the collective name for a series of management activities such as planning, implementation and process control of engineering projects. Modern engineering project management is characterised by complex content, dynamic process changes, long management cycle, etc. With the support of big data, it is easy to understand the data law, carry out intuitive management and meet the diversified management needs. Therefore, it is of great significance to design and develop an intelligent management system for the whole process of engineering projects based on big data. Engineering projects are different from the general factory batch production, each project is relatively independent, the project leader needs to fully coordinate resources, strengthen departmental communication and collaboration, through the project management platform, tracking and supervising the project progress, controlling the project cost and risk, in order to achieve the project goals and improve the quality of the project.

The new paradigm of digital collaborative management is beginning to take shape, and the use of digital technologies in EPC projects is an important part of national policy [1]. China's construction industry has also plunged into the field of intelligent buildings, and has achieved great results. Although digital EPC project management has a broad market prospect. However, the results are often unsatisfactory due to various factors. The new business model will be reconstructed, supported by digitalisation, to promote the development of the construction industry in the direction of intelligence and industrialisation. Finally, an integrated and collaborative management system based on in-depth design, data-driven production, visualised construction management, and intelligent decision-making will be formed to achieve the purpose of intelligent management [2]. In recent years, China has been vigorously promoting the development of EPC general contracting projects, and the basic projects represented by chemical construction projects have become the forerunners of EPC general contracting. With the increase in the number of EPC general contracting projects, the pressure of cost control continues to intensify, once the cost control is not appropriate, it will cause a huge waste of funds. The whole process cost control is a new cost control concept, which emphasises cost control from the whole process of EPC general contracting project to make up for the cost control gap and improve the cost management level. However, there are still some risk factors in the whole process of cost control in chemical construction projects, which restricts the advantage of whole process cost control. Based on this, it is of great practical significance to explore the whole process cost control measures of chemical construction project under EPC general contracting mode [3].

2 Cost control measures in the whole process of chemical construction project under EPC general contracting mode

2.1 Design Stage Cost Control

(1) Coordination of design with other phases

In the FI general contracting chemical construction project bidding stage, the general contractor should be from the objective layer and achieve the depth of all professional drawings construction drawings bidding, reasonable articulation of the design, kissing, construction, strictly in accordance with the time schedule out of the drawings, it is inevitable that the design and the schedule is not coordinated with the increase in costs caused by [4].

After winning the bid, the general contractor can according to the contract stipulates the construction period and the construction period nodes for the decomposition of the project progress plan. In the project progress plan decomposition at the same time, the general contractor can drawings review review Yang time as the basis for different professional, different special design progress plan adjustment, in strict accordance with the plan to promote the work [5].

(2) Design for Rational Use Limits

Limit design is the key to design cost control under F'C general contracting chemical construction project mode. Limit design requires designers and ensure that the construction project construction requirements to meet the premise, co-ordination of contractual agreements, the total project budget, the scope of the project, the design of construction drawings within the cost limit, and the use of appropriate economic indicators to estimate the price of the content of the drawings, comparison of the estimated price and the plan limit, tracking corrections to the design drawings, the estimated budget to control in the design limit [6].

(3) Scientifically optimised design

The general contractor's management can work with the person in charge of the design to analyse a wide range of engineering design options using the value engineering methodology, the working procedures of which are shown in Figure 1.

Considering that the chemical construction project consists of several parts, the scale is large, and the design scheme between each module is relatively uniform, the design of the process system scheme of syngas purification plant of a 1.8 million/year coal to

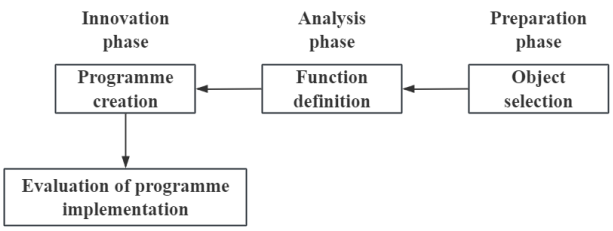

Figure 1: Working procedures for value engineering.

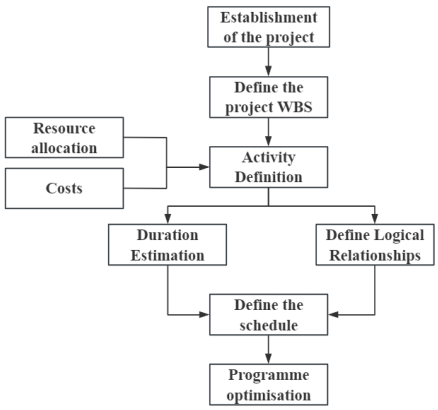

Figure 2: Traditional Project Scheduling Process.

ethylene glycol project can be taken as an example, and the value engineering method is applied to select the optimal scheme.

2.2 Elements of project schedule management

Project schedule management consists of two tasks: preparation and control of the schedule. Preparation of project schedule refers to the development of economically feasible schedule within a limited time! Project progress control refers to the implementation of the progress plan process, real-time monitoring of the actual progress, and compared with the progress plan to find deviations, identify the causes of deviation and develop measures to correct or adjust the initial plan, repeat this process until the completion of the project [7].

(1) Project schedule preparation

After the project schedule preparation is completed, the schedule should also be adjusted and optimised by taking into account other constraints such as project resources and costs. The traditional project schedule preparation process, see Figure 2.

(2) Project schedule control

With the deepening maturity of project management theory, project schedule management techniques have also been rapidly developed, and nowadays the more widely used schedule planning techniques include crosswalk diagrams, critical path method, plan review techniques, and critical chain method. Overview of the development of progress management techniques, as shown in Figure 3. One of the engineering applications are more crosswalk diagrams, critical path method, plan review technology critical chain method, the following is a brief introduction to these methods.

3 Deep learning based multi-target tracking approach

Multi-target tracking can be understood as a real-time state estimation problem for multiple variables. Assuming a total of n targets, the set of states of all targets in the current t-frame can be represented as $S_t = (S_t^1, S_t^2, \cdots, S_t^n)$, where the state of the ith target is represented as S_t^i. For the ith target, the sequence of states from its appearance to the end of the video stream is $S_{beginend}^i = \{S_{begin}^i, \cdots, S_{end}^i\}$. For all targets, the set of state sequences from their appearance to the end of the video stream in a total of t frames can be represented as $S_{1x} = (S_1, S_2, \cdots, S_t)$.

The most common current multi-target tracking framework is detection-based multi-target tracking, for detecting the state of the target in each frame S_t^i can be described by the detection results Det_t^i, i.e., the set of detection results for all targets in the tth frame can be expressed as $Det_t = (det_t^1, det_t^2, \ldots, det_t^n)$, Similarly, the set of detection results from the emergence to the end of a video stream with a total of t frames can be expressed as $Det_{1:t} = (det_1, det_2, \ldots, det_t)$. In summary, the multi-target tracking task needs to build a mathematical model in the all target state sequences to find the globally optimal solution through a mathematical model with maximum a posteriori probability, which can be expressed by Eq.1:

$$\hat{S}_{1.t} = arg_{s_{tr}} maxP(S_{1.t}|Det_{1.t}) \tag{1}$$

In this section, we will establish a mathematical model for the multi-target tracking task, and then analyse the current mainstream target detection methods, re-identification methods and trajectory prediction methods.

3.1 Target detection

(1) YOLOv5

The backbone network of YOLOv5 firstly uses the Focus structure to downsample the input image.The Focus structure obtains $\frac{640}{N}$ independent feature layers by taking a value every N pixels in the input 640×640 image, and then stacks the N feature layers, at this point, the information in the width and height dimensions is converted to the channel dimension, and then extracts the different features by means of convolution , as shown in Figure 4.

The backbone network following the Focus structure adopts the design style of the CSP network , with extensive use of CSP Blocks that have a CBL Block composition for stacking, where the CBL BlockCBL module consists of a convolutional layer, a Batch Normal (BN) layer and a LeakyReLU activation function for feature extraction. Unlike YOLOv4, which used the CSP structure only in the backbone network, in YOLOv5 the CSP structure is used throughout the entire network.There are two implementations of the CSP Block, the one in the backbone network is called CSP Block1 and the one in the back Neck part is called CSP Block2, the difference is that the former uses stacking in processing the feature Residual Units (Res-Unit) approach in the former and stacking CBL Blocks in the latter.

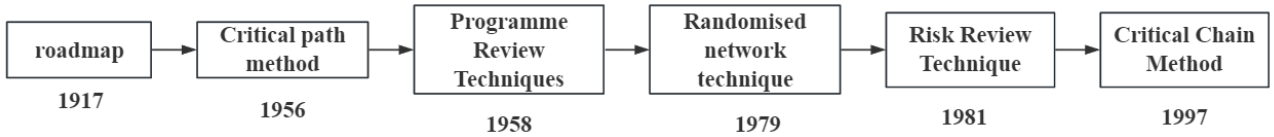

Figure 3: Overview of technical developments in progress planning.

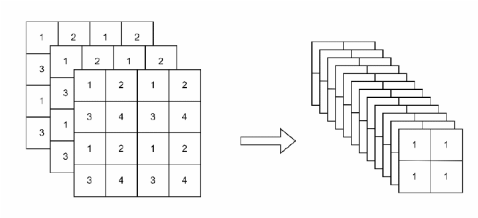

Figure 4: Focus Processing Logic Schematic.

The Neck part of YOLOv5 uses the PAFPN structure, which has become a design convention for the YOLO series of networks since YOLOv4, and the efficiency of feature fusion can be improved by the PAFPN structure, which will be analysed in detail in the paper.Both YOLOv4 and YOLOv5 use CIOU Loss as the loss function for detecting the target detection box (Bounding Box) loss function with the following expression:

$$L_{CIoU} = 1 - CIOU L_{CIoU} = 1 - CIOU \tag{2}$$

Of these, CIoU is:

$$CIoU = IoU - \frac{D_2^2}{D_2^2} - \alpha v \tag{3}$$

Where IoU is the intersection and concurrency ratio of the detection box to the real box, D_2^2 is the Euclidean distance from the centre point of the Bounding Box to the centre point of the Ground Truth, α is a balancing parameter that is not involved in the calculation of the gradient, and v is α parameter used to measure the consistency of the aspect ratio, and the expressions for α and v are, respectively:

$$\begin{cases} \alpha = \frac{v}{(1-IoU)+v} \\ v = \frac{4}{\pi}(arctan\frac{w^{gt}}{h^{gt}} - arctan\frac{w}{h})^2 \end{cases} \tag{4}$$

3.2 pedestrian re-identification

Re-ID based on metric learning usually uses a neural network to learn the similarity between two images to determine whether the input image matches an image in the retrieval library. This method can be viewed as an optimisation problem, and common loss functions are Contrastive Loss, Triplet Loss, Quadruplet Loss, etc. The contrast loss L_c is the most common loss function, noting $d_{I_a}^2, I_b$ as the Euclidean distance between the two input images, then its expression is as follows:

$$L_c = yd_{L_a,I_b}^2 + (1-y)\, max(\alpha - d_{L_a,I_b}, 0)^2 \tag{5}$$

where y denotes the input pedestrian label and $y = 1$ means that the two pedestrians belong to the same pedestrian, and vice versa for $y = 0$. $\alpha(0,1) \in$ is a threshold set according to the actual requirement.

Ternary loss requires input of three pictures, a fixed picture a (Anchor), a positive sample p (Postive) and a negative sample n (Negative), picture a and picture p form a positive sample pair and picture a and picture n form a negative sample pair. In addition the ternary loss defines a new threshold margin (denoted as α) such that the distance between two positive samples and the distance between negative samples of the same class is less than margin, this is done so that the features of the samples do not aggregate into a small space. The expression for the triad loss is given in the following equation:

$$L_t = max\left(d_{a,p} - d_{a,n} + \alpha, 0\right) \tag{6}$$

The idea of quaternion loss is similar to ternion loss, with the difference that it adds another negative sample image, i.e., there are two negative sample images in total. The expression for quaternion loss is:

$$L_t = \left(d_{a,p} - d_{a,m1} + \alpha\right) + \left(d_{a,p} - d_{a,n2} + \beta\right) \tag{7}$$

where α and β are thresholds set according to the actual situation, the former is called a strong push and the latter is called a weak push, and β is usually set to be less than α.

4 Big data platform design

4.1 Data collection layer

The data collection layer uses a Kafka cluster as a message queue to collect persistent data.The purpose of the Kafka cluster is to become a queuing platform, just use it to support both offline and online use of these two cases.Kafka collects data on disk at an access cost of (1). General data on disk is stored using BTree, access cost of O (logn), can effectively reduce disk IO, high throughput to meet the collection of most web logs, and Kafka is to show distributed system, producer-consumer agent can have multiple nodes, so that you can expand horizontally, in favour of the future expansion of the cluster to cope with the expansion of demand. The high Choropleth rate allows the system to process hundreds or thousands of messages per second even on an average node [8], which can satisfy most web log collection jobs. Explicit distribution assumes that the number of Producers, Brokers and Consumers are spread across multiple machines. Kafka supports parallel loading of data into Hadoop and real-time transfer to Spark streaming for quasi-real-time computation [9].As shown in Figure 5.

4.2 Data platform resource monitoring and statistical analysis layer

For Hadoop data platform to run stably and reasonably, it needs a system that can monitor and analyse the computing and storage

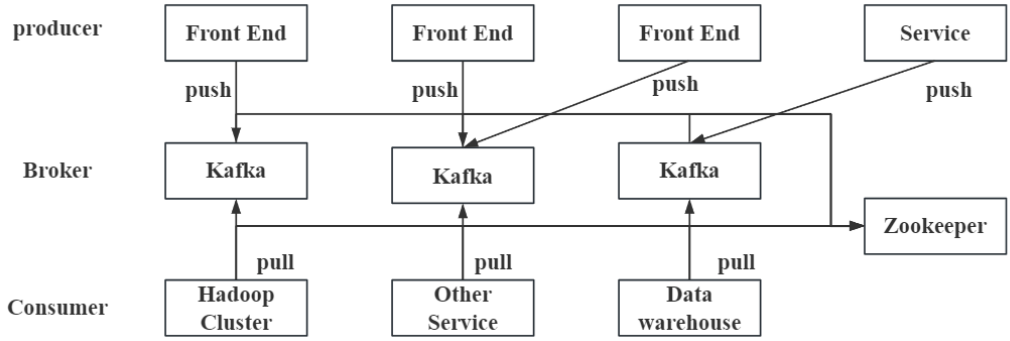

Figure 5: Kafka cluster schematic.

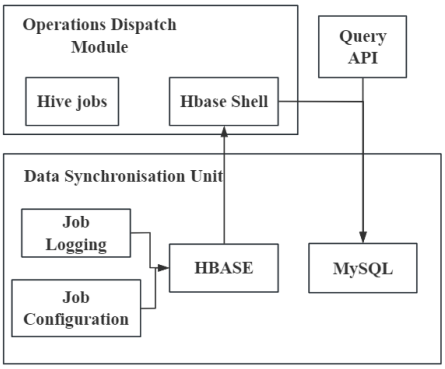

Figure 6: Overall Architecture Diagram of Computational Resource Statistics for Hadoop Data Platform Statistical Analysis System.

resources of the data platform, and this system can find out the problems and bottlenecks in the data platform in a very convenient way, so as to provide support and guarantee for the optimization and stable operation of the data platform. The data platform monitoring and statistical analysis layer is embedded into the Hadoop visualisation tool Hue as a module, and the JobAnanlvzer module is developed using Lue to provide the front-end display for the data platform monitoring and statistical analysis layer, and the back-end of the web display should be developed using Python because Hue uses the Diango web framework [10].

Figure 6 shows the overall architecture of the computing resource module of the statistical analysis system of the Hadoop big data platform. The original job logs and configuration information are saved in HDFS, and the logs and configuration information of the jobs are structured and saved in HBase through the data synchronisation unit provided by the system, and the job scheduling unit periodically checks the HBase and uses Hive to analyse and count the latest job information in HBase or directly using the HBase Shell. The job scheduling unit periodically checks HBase and saves the latest job information in HBase using Hive or directly using the HBase Shell to analyse the statistics, and finally saves the data of the completed statistical analysis in MYSQL for access by the data query API.

5 The application value

The platform can be used in construction projects, municipal infrastructure projects, rail transit projects, water conservancy and water projects and other engineering fields. In view of the shortcomings of traditional project management, such as information closure and information asymmetry, the establishment of block 'alliance chain' based on the industrial Internet platform allows each participant in the construction not only to perform its own duties, but also to supervise and collaborate together, which is of far-reaching significance for the creation of an open and transparent, mutually beneficial and helpful, and information-sharing and even benefit-sharing operating environment in the construction industry. This is of far-reaching significance for the creation of an open and transparent operating environment for the construction industry with mutual benefit and assistance, information sharing and even benefit sharing.

Founded in 2000, a company is committed to the whole life cycle of construction projects, the whole process of consulting services, the company's 'smart city of the future' as the corporate mission, the first in the country to put forward the 'Digital Intelligence +' consulting services model, independent research and development of the 'Digital Intelligence +' consulting services. The company is the first in China to propose the 'Digital Intelligence+' consulting service model, and has independently developed the 'Digital Intelligence+' consulting platform 'Consulting and Construction Cloud', which is a deep fusion of digital intelligence and engineering consulting, and empowers the users.

The company has the qualification of Class A engineering consulting, comprehensive qualification of engineering supervision, Class A qualification of engineering cost, Class A qualification of bidding agent, Class A qualification of project management and whole process engineering consulting in Sichuan Province, Class B qualification of cultural relics preservation supervision, and Class B qualification of water conservancy project, and it is a double-entry unit of PPP consulting institution library of the Ministry of Finance and the Development and Reform Commission, and one of the first batch of pilot units of whole process engineering consulting of the

Department of Construction and Residence of Sichuan Province, and it has participated in the compilation of a number of national standards and provincial standards. The company has participated in the compilation of a number of national and provincial standards.

The company's industrial chain covers investment decision-making consulting, project management, construction supervision, engineering cost consulting, bidding and procurement consulting, survey and design management consulting, BIM consulting and other engineering consulting services, and provides engineering consulting enterprises with digital intelligent construction solutions.

So far, the company has been contracted to provide consulting services for construction projects with a total investment of RMB 500 billion, and has won more than 200 honours such as National Luban Prize, National Quality Engineering, China Steel Structure Gold Award, Tianfu Cup, Furong Cup, Excellent Engineering Cost Consulting Achievement, Provincial and Municipal Quality Engineering, Provincial and Municipal Standard Chemical Site, Provincial Excellent Consulting Achievement, etc., and has obtained a number of patents and softwritings.

In the era of digital economy of engineering consulting industry, the company will take scientific and technological innovation as the forerunner, continuously iterating 'Consulting and Construction Cloud' platform, integrating the most cutting-edge intelligent construction technology and equipment, and creating value for the customers with continuously iterating 'Digital Intelligence +' engineering consulting service.

6 Conclusion

In summary, EPC mode is a widely used mode in the engineering contracting market, and project cost management is the guarantee for the effective application of this mode. Under the EPC mode, the design stage, procurement stage and construction stage contain many cost influencing factors, therefore, the general contractor should coordinate the design and other stages, focus on the limit design, reasonably design the procurement plan, and select the best suppliers. At the same time, the general contractor should pay attention to the management of subcontracting units, pay attention to the time of project payment, and control the safety risk factors of construction to ensure the effect of the whole process of cost control. This paper is based on the research background of big data on the project management of new research and combination. In the development of big data today, the project management platform combined with big data will get better development and more scientific management data, this paper has a greater significance in the study of project combination.

At present, a number of domestic organisations have launched intelligent management platforms for the whole construction process. For example, the intelligent construction information management platform researched and developed by China Construction First Bureau is applied to Shenzhen International Convention and Exhibition Centre, which realizes intelligent management on 11 management systems, such as personnel and machinery positioning, material tracking and acceptance, progress control, quality inspection, TSP environment monitoring, etc., through the 'Internet +' intelligent collaboration platform, constituting the CPU of the whole intelligent construction site, making the construction process more intelligent and efficient. CPU, making the construction process of Shenzhen International Convention and Exhibition Centre project more intelligent and efficient. The comprehensive digital management platform for the whole process of infrastructure developed by State Grid has digital, intelligent and open sharing features, and serves the 'four-in-one' intelligent platform (digital management platform, infrastructure work platform, resource sharing platform and decision-making support platform) for all levels of grid infrastructure professions and related professional users, which realises the sharing of data of all infrastructure professions, and the sharing of data of all departments of the company. It realises the sharing of professional data of infrastructure, the business integration of various departments of the company and the synergistic development of the company and related industries.

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